

2-Mark Questions & Answers MPCA Unit 3: The Intel Microprocessor 8086 Architecture

Q1. List the functional units of the 8086 microprocessor.

The 8086 CPU is divided into two independent functional units:

1. **B**
address

Q2. What is memory segmentation in 8086?

Memory Segmentation is a technique that divides the 1MB physical memory space into 64KB logical blocks called segments (Code Segment, Data Segment, Stack Segment, Extra Segment). It allows a 16-bit processor to address 20-bit physical addresses using segment base registers and offset registers.

Q3. Define minimum mode and maximum mode of 8086.

- **Minimum Mode** ($\overline{\text{MN}}/\text{MX}} = 1$): Single-processor system mode where 8086 generates all bus control signals ($\overline{\text{RD}}$, $\overline{\text{WR}}$, $\overline{\text{ALE}}$, $\overline{\text{M/IO}}$) directly.

- **N**
by an

Q4. What is the purpose of demultiplexing the address/data bus?

The 8086 uses multiplexed pins AD_{0-15} to save physical pin count. **Demultiplexing** separates the 20-bit memory address from 16-bit data using octal latches (74LS373) driven by the Address Latch Enable ($\overline{\text{ALE}}$) signal.

Q5. State the size of the 8086 instruction queue.

The 8086 Bus Interface Unit (BIU) contains a **6-byte Instruction Queue** that prefetches instruction bytes sequentially from memory when the EU is busy, enabling instruction pipelined prefetching.